

Harry Guan

✉ hguan.trading@gmail.com | 📞 (669) 279 8250 | 📧 ~harryguan | 🌐 hguan-dev

EDUCATION

Northwestern University

B.A. in Mathematics and Computer Science, B.M. in Trombone Performance

Expected December 2027

Evanston, IL

- **Cumulative GPA:** 3.8/4.0 | **Major GPA:** 3.9/4.0 | **Dean's List:** 8/8 Quarters
- **Relevant Coursework:** Probability and Stochastic Processes, Derivatives Theory, Regression Analysis, Machine Learning, Deep Reinforcement Learning, Operating Systems, Parallel Computing, Linear Algebra, Design and Analysis of Algorithms

EXPERIENCE

IMC Trading

Incoming Quantitative Trading Intern

June 2026 – August 2026

Chicago, IL

PEAK6 Capital Management

Incoming Quantitative Trading Intern

May 2026 – June 2026

Chicago, IL

LinkedIn

Machine Learning Engineer Intern - Ads

June 2025 – September 2025

Mountain View, CA

- Engineered a **Dirichlet**-parameterizing deep CTR model with **DCNv2** interaction towers, optimizing a digamma-based loss to output α, β parameters for **Thompson Sampling** and predict full click-type distributions versus singular scalar output
- Implemented **clipped inverse propensity weighting** to mitigate selection bias and applied **isotonic calibration** across chargeability and click-type heads, ensuring strict probability alignment for multiple ad objectives
- Designed vectorized murmur-hash input pipelines and instrumented per-head ranking metrics, improving inference accuracy to **reduce loss by 17%** and driving a **\$1.2M+ increase** in annualized advertisement revenue

IMC Trading

Launchpad - Quantitative Trading Cohort

May 2025

Chicago, IL

- Secured **1st place** out of 30 in IMC Trading's futures market-making competition by implementing dynamic bid-ask spread sizing using the **Avellaneda-Stoikov** model and data-feed pipeline monitoring to capture order-flow edge
- Engaged in intensive lectures and simulations covering **options pricing, futures mechanics**, market microstructure, and overall quantitative trading strategies, with hands-on applications in **risk management** and position sizing

Susquehanna International Group

Discovery Day - Quantitative Trading Cohort

April 2025

Bala Cynwyd, PA

- Achieved the **highest PnL** in the trading cohort of 40 by engineering a pandas-based batch processing pipeline to analyze **30,000+ BTC order book records** across multiple exchanges, identifying arbitrage opportunities and securing profit

PROJECTS

Texas Hold'em Poker Solver ([GitHub](#))

August 2025 – December 2025

- Engineered a high-throughput **Monte Carlo CFR** C++ solver to approximate Nash Equilibria via iterative self-play, implementing **zero-allocation** hot paths to generate **2.2M+** ground-truth samples at **>50,000 iterations/sec**
- Designed an XGBoost and PyTorch pipeline to predict optimal mixed-strategy probability distributions from 58 engineered features, achieving a **0.274 KL Divergence** and **0.206 Total Variation Distance** to compress the solution space for real-time inference

NU FinTech Club Trading Competition ([GitHub](#))

June 2025 – September 2025

- Advanced a C++ **exchange simulator** to support dynamic market scenarios, including manual trading and multi-exchange arbitrage; researching current market structure to inform order book and participant design
- Engineered Python trading agents, including sporadic signal-based traders and participants placing information-rich large orders, to simulate complex market dynamics and test algorithm robustness in volatile exchange conditions

HONORS AND AWARDS

IMC Trading Market-Making Competition | *1st Place Overall*

Susquehanna International Group Arbitrage Competition | *1st Place Overall*

American Invitational Mathematics Examination Qualifier (4x) | *Top 5% in the American Mathematics Competition*

IMC Trading Low Latency Competition | *2nd Place Overall*

Northwestern University Algorithmic Trading Competition | *2nd Place Cryptocurrency Exchange*

USA Coding Olympiad | *Gold Division, Top 7% in Contestants*

ADDITIONAL

Programming Languages: Python, C++, C, Golang, Rust, Java, TypeScript, Bash, x86 Assembly

Frameworks/Libraries: NumPy, TensorFlow, Pandas, PyTorch, scikit-learn, PyTest, PyBind, GTest, Node.js, Matplotlib

Infrastructure: UNIX, CUDA, Amazon Web Services, GCC, OpenMP, Nginx, Jenkins, Docker, Git, GitHub + Actions

Interests: Texas Hold'em Poker, Teamfight Tactics, League of Legends, Classical Music, Orchestral Conducting, Tarantino Movies